

Workbook: Articulations

Week 2 workbook · BIO 004 Human Anatomy · Dr. Sharilyn Rennie · Solano Community College

A comprehensive workbook on the structural and functional classifications of joints, the anatomy of synovial joints, the six types of synovial joints, joint movements, and key joints of the body. Includes clinical anatomy for sprains, dislocations, and common joint injuries.

How to use this workbook: Read each question carefully and write your answer in the space provided. Work without looking up the answer first. Once you have written your answer, check it against the online workbook by clicking *Reveal* on that question. Star or circle anything you missed and come back to it later in the week.

Joint classifications · 4 questions

Q01 · DOK 1 · Recall

Name the three structural classifications of joints and what tissue holds each one together.

Q02 · DOK 1 · Recall

Name the three functional classifications of joints and how much movement each allows.

Q03 · DOK 1 · Recall

Name the three types of fibrous joints with an example of each.

Q04 · DOK 1 · Recall

Name the two types of cartilaginous joints and give an example of each.

Synovial joint anatomy · 6 questions

Q05 · DOK 1 · Recall

Name the parts of a typical synovial joint.

Q06 · DOK 1 · Recall

What is synovial fluid, what produces it, and what does it do?

Q07 · DOK 1 · Recall

What is articular cartilage, what kind of tissue is it, and why does joint damage to it heal poorly?

Q08 · DOK 1 · Recall

What are menisci, where are they found, and what do they do?

Q09 · DOK 1 · Recall

What is a bursa, where is it found, and what does it do?

Q10 · DOK 1 · Recall

What is a tendon sheath, and how is it like a bursa?

Six types of synovial joints • 6 questions

Q11 • DOK 1 • Recall

Describe a plane (gliding) joint and give one example.

Q12 • DOK 1 • Recall

Describe a hinge joint and give two examples.

Q13 • DOK 1 • Recall

Describe a pivot joint and give two examples.

Q14 • DOK 1 • Recall

Describe a condyloid (ellipsoid) joint and give an example.

Q15 • DOK 1 • Recall

Describe a saddle joint and give the classic example.

Q16 • DOK 1 • Recall

Describe a ball-and-socket joint and give the two main examples.

Joint movements · 8 questions

Q17 · DOK 1 · Recall

Define flexion and extension, and give an example.

Q18 · DOK 1 · Recall

Define abduction and adduction.

Q19 · DOK 1 · Recall

Define medial and lateral rotation.

Q20 · DOK 1 · Recall

Define circumduction.

Q21 · DOK 1 · Recall

Define pronation and supination of the forearm.

Q22 · DOK 1 · Recall

Define dorsiflexion and plantar flexion of the ankle.

Q23 · DOK 1 · Recall

Define inversion and eversion of the foot.

Q24 · DOK 1 · Recall

Define protraction and retraction.

Major joints · 5 questions**Q25 · DOK 2 · Apply**

The shoulder is the most mobile joint in the body. What anatomical features make it so mobile, and what is the trade-off?

Q26 · DOK 2 · Apply

Compare the hip joint with the shoulder joint by structure and the trade-off between motion and stability.

Q27 · DOK 2 · Apply

The knee is described as a modified hinge joint. Why 'modified', and what major intra-articular structures stabilize it?

Q28 · DOK 1 · Recall

What kind of joint is the elbow, and what motions does it allow?

Q29 · DOK 1 · Recall

What kind of joint is the temporomandibular joint, and what unusual feature does it have?

Clinical reasoning · 3 questions

Q30 · DOK 3 · Reason

A sprain is an injury to a ligament. Using what you know about ligament structure, predict why severe sprains heal slowly.

The anterior shoulder is the most common dislocation site in the body. Using the joint's anatomy, explain why anterior dislocations are common.

Osteoarthritis (OA) and rheumatoid arthritis (RA) both damage synovial joints. Compare what each one attacks and predict how the joint cavity looks in each.